

SEC P vs NP — Frameworks (live)

SEC (autonomous) — attributed to Ludovico Kubler

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Summary table

Framework ID	Status	Fitness	Generation	Parent	Name
fw_b9e7d103d0	ELABORATING	0.000	0	-	Ergodic Circuit Framework

Details

Ergodic Circuit Framework (fw_b9e7d103d0)

- **Status:** ELABORATING
- **Fitness:** 0.000
- **Taxonomy:** COMM_COMPLEXITY
- **Target invariant:** communication_entropy_barrier (real number ≥ 0) $\rightarrow$ bounds The minimum communication complexity in a k-party number-on-forehead model, where lower bounds are derived from the growth rate of Kolmogorov-Sinai entropy in associated dynamical circuits. Aims to prove $\omega(\log n)$ lower bounds for explicit functions like Disjointness, avoiding natural proofs via non-uniform dynamics.

Primitives: - measurable_dynamical_circuit (tuple (C, X, μ, T)): A Boolean circuit C where gates are labeled by operations over a probability space (X, μ) , and $T: X \rightarrow X$ is a measure-preserving transformation
- orbit_signature (function: $N \rightarrow [0,1]$): For a fixed input $x \in \{0,1\}^n$, the orbit signature is the sequence $(\mu(C(T^k(x))))_k$, capturing how the circuit's output evolves under repeated application
- cross_correlation_flow (matrix $\in R^{d \times d}$): For a depth- d circuit, this matrix $F_{i,j} = \int |C_i(x) - C_i(T^j(x))|^2 d\mu(x)$ measures how perturbations via T propagate through layers.
Computable via

Tentative axioms: - A1: Any circuit with $o(n)$ communication complexity induces a dynamical system with sublinear Kolmogorov-Sinai entropy growth. - A2: Measure-preserving transformations corresponding to AC^0 circuits have bounded mixing time, while those for PSPACE-complete computations exhibit super-polynomial mixer-profile decay. - A3: If a family of circuits computes a function with high cross_correlation_flow norm, then its communication complexity is $\Omega(n^\delta)$ for some $\delta > 0$.